

Paper A-03: Earth Ocean Currents: A Constraint-Driven Flux Dynamics (CDFD) Perspective

Steve Bico Mujjabi, MD

Independent Researcher

ORCID: 0009-0001-0556-5516

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Abstract

This paper treats ocean currents as a CDFD/AFL modelling hypothesis. Wind stress, buoyancy forcing, heat, salt, and mass transport are read as fluxes Φ moving through basin geometry, stratification, bathymetry, mixing limits, and freshwater constraints C . The claim is not that CDFD/AFL replaces physical oceanography. The claim is that the notation can organise measurable questions about transport, bottlenecks, responsiveness, and long memory in the ocean system.

1 Series Context

Part I sets the flow–constraint language used here [1]. Part II develops the Life Number and the origins-of-life use of the same notation [2]. Part III applies AFL to biology and medicine [3]. Part IV uses the language as a cross-domain stress test, with measurement kept separate from analogy.

Notation. In this paper, C names the active ocean constraint or capacity burden. Φ is the driving transport, S is the ocean response, and M_s is retained structural memory. Λ is mentioned only where the Part II Life Number is being referenced [2].

2 Ocean Transport Frame

The ocean stores and redistributes heat, salt, carbon, nutrients, and momentum. Its flow is constrained by basin walls, seafloor geometry, density structure, wind forcing, sea ice, freshwater input, and mixing depth. In the CDFD/AFL reading, a current system is not only a velocity field; it is a flow path with capacity, inertia, adaptive response, and memory.

CDFD term	Ocean-current reading	Example proxy
Φ	ocean transport or forcing	Sverdrup transport, heat flux, freshwater flux
C	active constraint	stratification, bathymetry, sill depth, friction
S	responsiveness	mixing, eddy compensation, current-path adjustment
M_s	structural memory	heat storage, salinity anomaly, overturning inertia
Ψ_s	operating ratio	balanced, constrained, stalled, or amplified transport

3 Candidate Variable Map

4 Memory and Circulation Change

Ocean memory is central to the mapping. Heat stored below the mixed layer, persistent salinity anomalies, ice-ocean feedback, and overturning inertia can make present circulation depend on prior forcing. In CDFD/AFL terms, high M_s means the current state carries a record of earlier flux and constraint. That does not by itself prove a new mechanism; it gives a way to state what must be measured.

5 Measurement Layer

A useful ocean application specifies the current system, the forcing window, the proxy for Φ , the constraint term C , the response term S , and the memory term M_s . Candidate failure conditions include persistent transport weakening, blocked heat redistribution, salinity-driven stratification, or a shift in overturning regime. The mapping has value only if it improves measurement or interpretation beyond ordinary oceanographic diagnostics.

6 Current Runtime Diagnostic

The Part E diagnostic run includes an ocean-transport adapter row and the universal cascade stress test. These outputs check finite runtime behavior in the current CDFD Runtime [4]. They do not validate any ocean claim; they record model behavior that later data work can test.

7 Tests That Can Break the Mapping

The ocean-current mapping weakens if the chosen proxies for flux, constraint, responsiveness, and memory do not track observed transport changes; if a standard ocean model explains the same behavior more cleanly; or if the CDFD terms cannot be tied to reproducible ocean data.

8 Conclusion

Ocean currents are a natural test case for flow, constraint, response, and memory. CDFD/AFL is useful here only as a disciplined measurement language. The release-level claim remains limited to that role.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Release-local runtime diagnostics are available under each Part's `outputs/` folder, with release-wide synthesis artifacts under `Part_E.Synthesis/supplementary/`, `Part_E.Synthesis/outputs` and `Part_E.Synthesis/figures/`.

9 Reference Layer

References

- [1] Steve Bico Mujjabi. Cdfd part i: Fundamental physics - constraint-driven field theory and flux dynamics, 2026.
- [2] Steve Bico Mujjabi. Cdfd part ii: Origins of life and tri-regime bioenergetics - constraint-driven flux dynamics, 2026.
- [3] Steve Bico Mujjabi. Cdfd part iii: Adaptive flux limitation biology and medicine - constraint-driven flux dynamics, 2026.
- [4] Steve Bico Mujjabi. Cdfd runtime: Constraint-driven flux dynamics and cdfl execution engine, 2026.